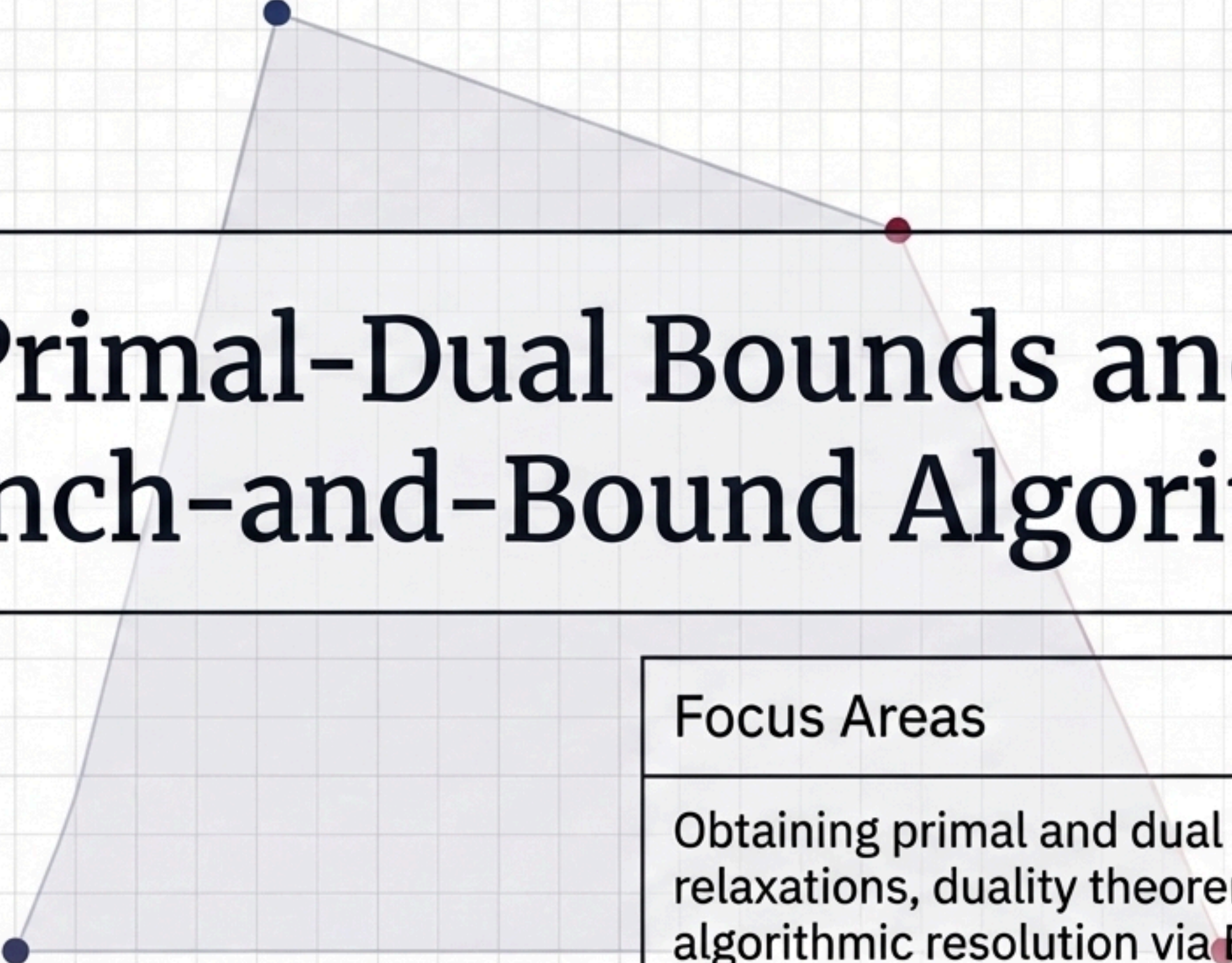


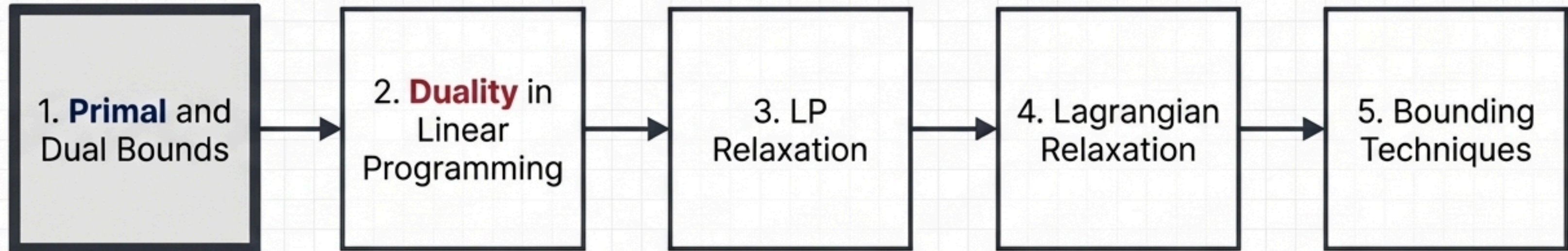


Primal-Dual Bounds and Branch-and-Bound Algorithm

Focus Areas

Obtaining primal and dual bounds, LP relaxations, duality theorems, and exact algorithmic resolution via Branch-and-Bound.

Motivation: Before attempting to solve complex combinatorial problems, we must define the mathematical limits of our optimal objective values.



Primal and Dual Bounds

Primal Bound

A valid upper limit (in minimization) on the optimal objective value.

Obtained by evaluating the objective function at any feasible solution.

The true minimum is $\leq$ this bound.

$$z_{opt} \leq z_{primal}$$

Dual Bound

A valid lower limit (in minimization) on the optimal objective value.

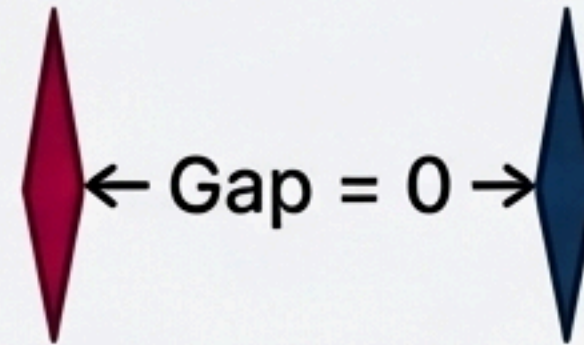
Found for example via LP Relaxation of integer program (e.g., ignoring integrality constraints in IP, MILP).

Always optimistic because relaxations expand the feasible region or are as good as integer.

$$z_{dual} \leq z_{opt}$$

Motivation: Having defined the bounds, we must establish how to measure the distance between them to mathematically guarantee an optimal solution.

Primal Bound and Dual Bound



The Optimality Gap

The absolute or relative difference between the best known primal bound (best feasible solution found) and the best known dual bound (theoretical limit).

Certifying Optimality

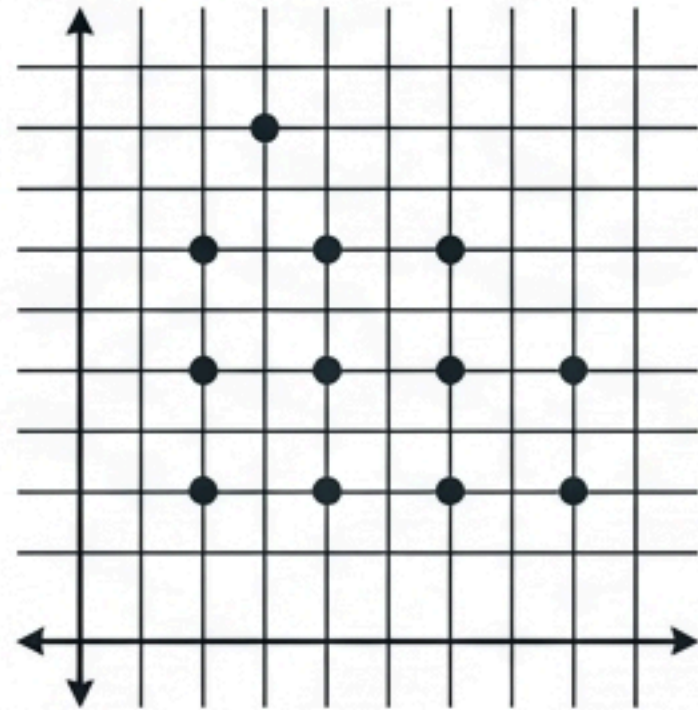
Optimality is strictly certified when the Optimality Gap equals zero (Primal Bound = Dual Bound).

Implication

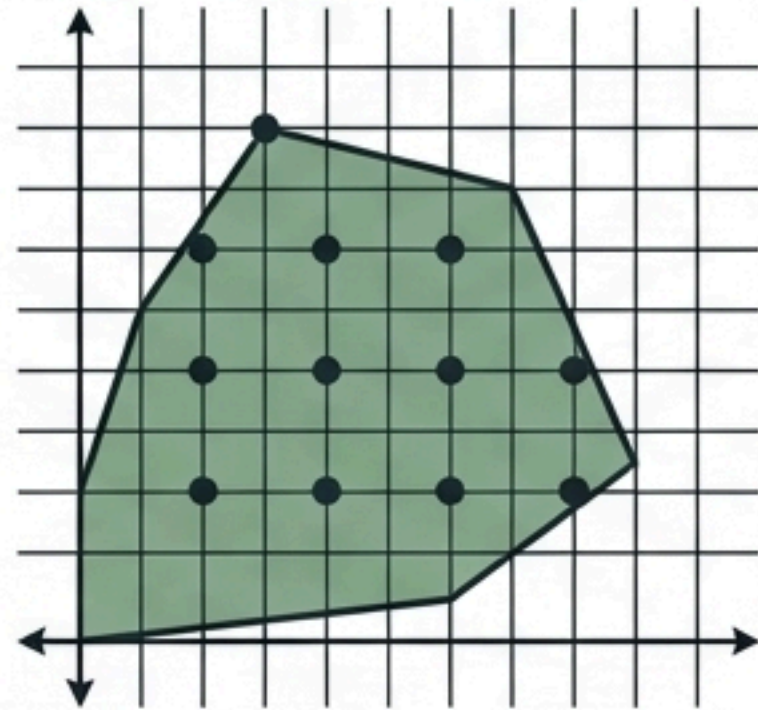
Proves no better solution can possibly exist within the feasible space.

Motivation: Let us apply this theoretical relaxation to a concrete mathematical model to extract numerical bounds.

LP Relaxation Definition: The linear programming problem obtained by dropping integrality constraints from an Integer Program (IP) or Mixed-Integer Linear Program (MILP).



Integer Program (IP)
Feasible Region (Discrete)



LP Relaxation
Feasible Region (Continuous)

LP Relaxation and Finding a Dual Bound for IP/MILP

Explanation: LP Relaxation expands the feasible region by allowing non-integer values. For a maximization problem (as shown), the optimal solution of the relaxed problem (continuous) must be greater than or equal to the optimal solution of the original integer problem. This creates a valid upper (dual) bound. For minimization, it provides a valid lower (dual) bound. This bound is essential for bounding the search space.

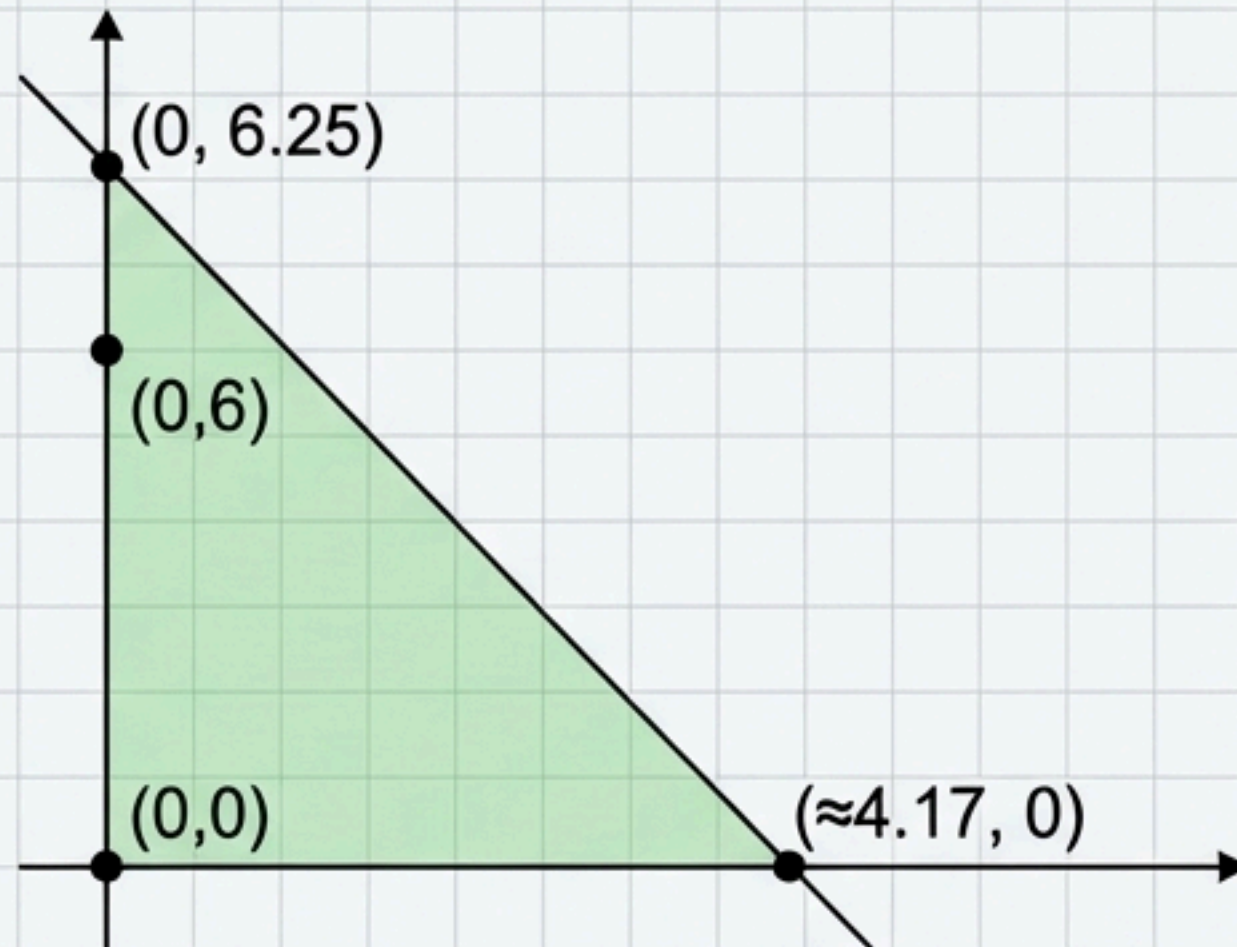
Mechanism: Allows variables to take continuous fractional values instead of strict integers.

Purpose: This method guarantees a valid bound for our solution and the problem can be solved with a simplex method easily.

Example (IP problem)

Problem

$$\begin{array}{ll}\max & 5x_1 + 4x_2 \\ \text{subject to} & 6x_1 + 4x_2 \leq 25 \\ & x_1, x_2 \in \mathbb{Z}_{\geq 0}\end{array}$$



Objective

Solve the LP relaxation geometrically to find dual bounds.

Components & Step-by-Step Resolution

1. **Relaxation:** Drop $Z \geq 0$, assume continuous bounds $x_1, x_2 \geq 0$.
2. **Vertices Evaluation:** $(0, 0) \Rightarrow Z = 0$; $(25/6, 0) \Rightarrow Z \approx 20.83$; $(0, 25/4) \Rightarrow Z = 25$.
3. **Dual Bound:** Continuous maximum $Z_{LP} = 25$. (Valid upper bound for maximization).
4. **Primal Bound:** The continuous solution $(0, 6.25)$ is not integer. The best feasible integer solution found is $(0, 6)$. Primal bound = 24.
5. **Optimality Gap:** $|25 - 24| = 1$. The gap is not zero, so optimality is not yet certified.

Duality in Linear Programming

The Dual Problem

Every linear program (primal) has an associated dual. Primal variables become dual constraints; primal constraints become dual variables.

Weak Duality Theorem

The primal objective is bounded by the dual objective (e.g., $c^T x \leq b^T y$ for primal maximization). Any feasible dual solution provides a dual bound.

Strong Duality Theorem

If either problem has a finite optimal solution, both do, and their optimal objective values are strictly equal ($Z^* = W^*$).

Complementary Slackness

Definition of Slackness: For a constraint (e.g., $ax \leq b$), slackness is the difference between the RHS and LHS (slack = $b - ax$). For a variable ($x \geq 0$), slackness is its value.

Explanation: At optimality, if a primal constraint has non-zero slack (is not binding), its corresponding dual variable must be zero. Conversely, if a primal variable is non-zero, its corresponding dual constraint must have zero slack (be binding). This relationship is symmetric.

Dual Problem Formulation

Complementary Slackness & Primal-Dual Relationship

At optimality, **active** constraints in the primal (zero slack) imply the corresponding dual variables are non-zero. **Non-active** primal constraints (non-zero slack) imply the corresponding **dual variables are zero**. The primal objective coefficients become the **right-hand side** of the dual constraints, and the primal right-hand side values become the dual objective coefficients.

Primal Problem Formulation:

$$\begin{aligned} \max Z &= 3x_1 + 2x_2 \\ \text{subject to } x_1 + x_2 &\leq 4, \\ 2x_1 + x_2 &\leq 5, \\ x_1, x_2 &\geq 0 \end{aligned}$$

Objective:
Write the
dual problem.

Dual Components Formulated:

$$\begin{aligned} \min W &= 4y_1 + 5y_2 \\ \text{subject to } y_1 + 2y_2 &\geq 3, \\ y_1 + y_2 &\geq 2, \\ y_1, y_2 &\geq 0 \end{aligned}$$

Primal Problem Solved, Dual Problem Solved

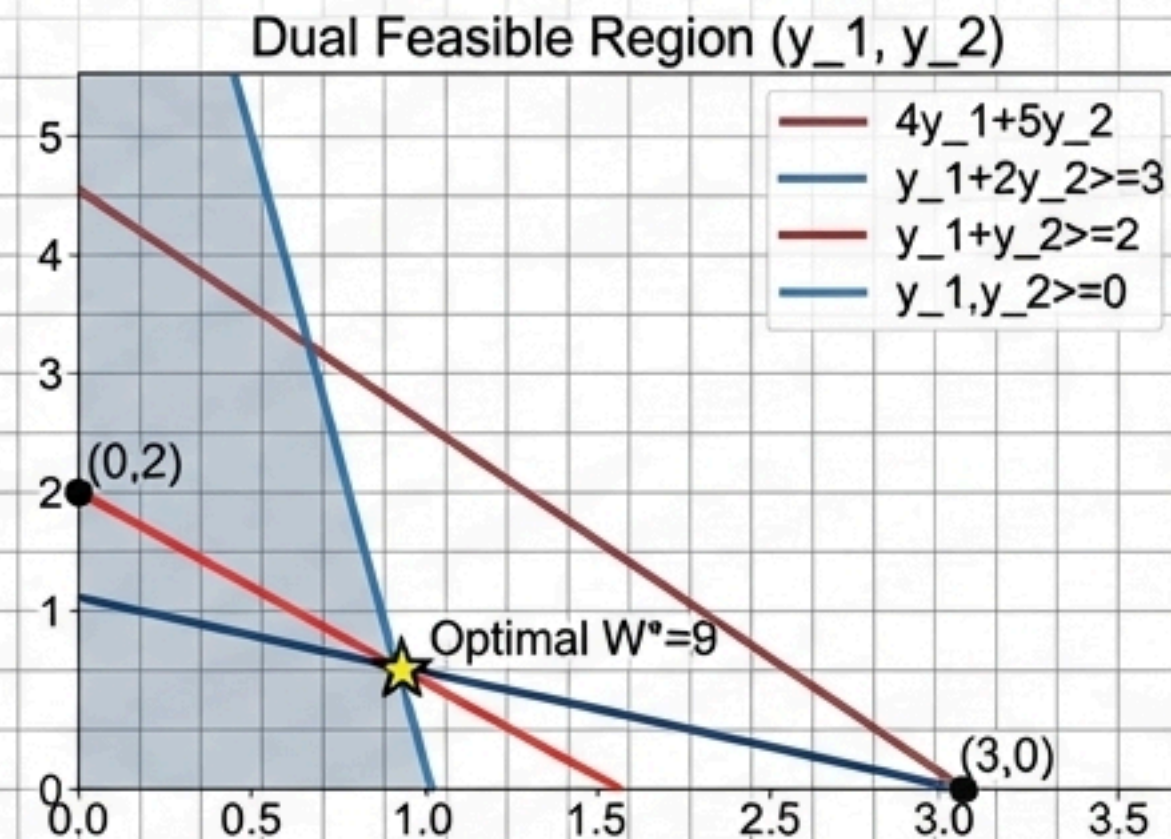
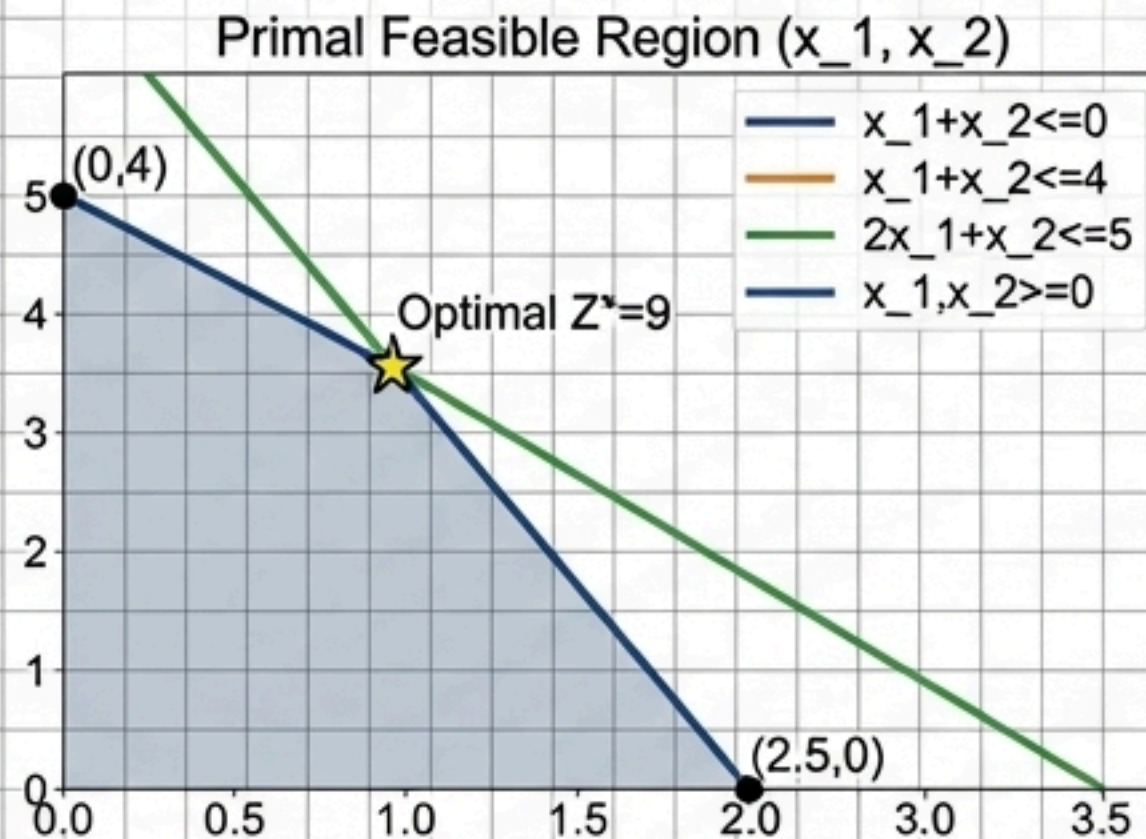
Primal Solve

Intersection of constraints at (1, 3).
Vertices evaluated:
 $Z(0,4) = 8$, $Z(1,3) = 9$, $Z(2.5,0) = 7.5$.
Maximum $Z^* = 9$ at $(x_1=1, x_2=3)$.

Dual Solve

Intersection of constraints at (1, 1).
Vertices evaluated:
 $W(0,2) = 10$, $W(1,1) = 9$, $W(3,0) = 12$.
Minimum $W^* = 9$ at $(y_1=1, y_2=1)$.

Graphical Solution Visualization



Strong Duality Verification

The optimal primal objective value ($Z^* = 9$) exactly equals the optimal dual objective value ($W^* = 9$).

Bounding Techniques for Optimization

Current Section:

1. Primal and Dual Bounds

2. LP Relaxation

3. Duality in Linear Programming

4. Bounding Techniques

5. Branch and Bound: An Exact Algorithm

Approach	Use Case	Bound Quality	Properties
LP Relaxation	IP, MILP	Variable	Fast to compute, drops integrality constraints.
Lagrangian Relaxation	Constrained IP	Strong	Hard constraints moved to objective function and penalized by multipliers.
Surrogate Relaxation	IP	Moderate to Strong	Aggregates multiple constraints into a single weighted row.

Lagrangian Techniques

Motivation: To handle difficult constraints, we can relax them and incorporate them as penalties in the objective function, creating a simpler problem to solve.

Lagrangian Relaxation Definition

Consider an Integer Program (IP):

$$\begin{aligned} \max z &= c^T x \\ \text{s.t. } Ax &\leq b \text{ (hard constraints)} \\ Dx &\leq e \text{ (easy constraints)} \\ x &\in \mathbb{Z}_+^n \end{aligned}$$

Introduce non-negative Lagrange multipliers $\lambda \geq 0$ for the hard constraints. The Lagrangian Relaxation $L(\lambda)$ is:

$$\begin{aligned} L(\lambda) &= \max_x (c^T x + \lambda^T (b - Ax)) \\ \text{s.t. } Dx &\leq e, x \in \mathbb{Z}_+^n \end{aligned}$$

How it Works (Explanation)

1. The hard constraints $Ax \leq b$ are removed from the constraint set.
2. They are moved to the objective function as a penalty term $\lambda^T (b - Ax)$.
3. If a solution x violates $Ax \leq b$, the term $\lambda^T (b - Ax)$ will be negative, reducing the objective.
4. For any $\lambda \geq 0$, the optimal value $L(\lambda)$ is an upper bound on the original IP's optimal value z^* .
5. The goal is to find the best (minimum) upper bound:
 $\min_{\lambda \geq 0} L(\lambda)$.

Examples (Without Solutions)

Example 1:

$$\begin{aligned} \max & 5x_1 + 8x_2 \\ \text{s.t. } & x_1 + x_2 \leq 1 \text{ (relax with } \lambda) \\ & 2x_1 + 3x_2 \leq 4 \\ & x_1, x_2 \in \{0, 1\} \end{aligned}$$

Relaxed Objective:

$$\max (5x_1 + 8x_2) + \lambda(1 - x_1 - x_2)$$

Example 2 (TSP):

Relaxing the degree constraints (each city visited exactly once) in the Traveling Salesperson Problem formulation, leaving only the subtour elimination constraints.

Exact Algorithm Framework

Exact Algorithm Framework: Guarantees finding the **absolute mathematical optimum** and proving its optimality in finite time. Utilized for strict safety or financial constraints.

Main Idea: "Divide and Conquer" - Partitioning the feasible region.

Pillar 1: Branching

Systematically partitioning the feasible region into smaller subproblems.

Pillar 2: Bounding

Utilizing relaxations to compute theoretical limits for each subproblem.

Pillar 3: Fathoming (Pruning)

Permanently closing a node if its LP is infeasible, or its LP solution is already strictly integer.

Motivation: With the exact algorithm fully defined, we apply it to a final mathematical program to observe the bounding and fathoming mechanics in real time.

Mathematical Logic

Formulation: $\max Z \text{ s.t. } x \in S$

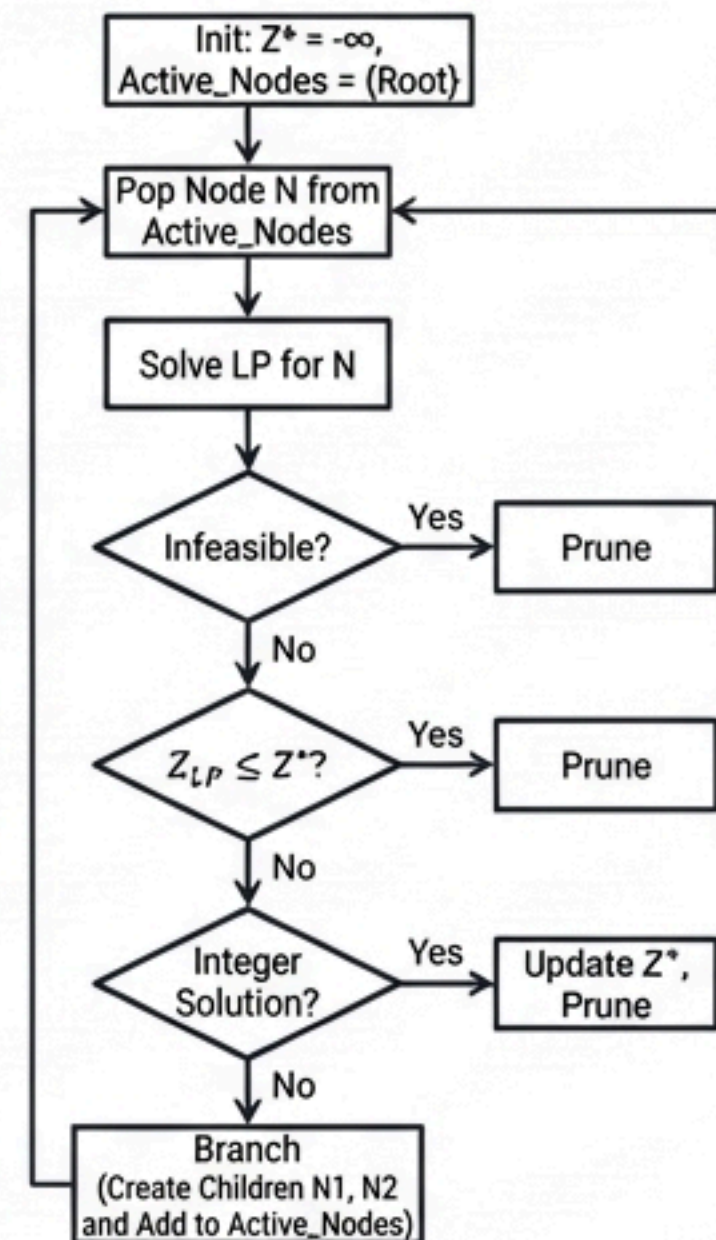
Branching Constraints:

$$x_i \leq \lfloor v \rfloor \text{ and } x_i \geq \lceil v \rceil$$

Optimality Fathoming:

$$Z_{LP} \leq Z^*$$

Flowchart



Structured Pseudocode

```
Best_Z = -infinity;
Active_Nodes = [Root Node];
While Active_Nodes not empty:
    Current = Select(Active_Nodes);
    Remove Current from Active_Nodes;
    LP_Sol, Z_LP = Solve_LP(Current);
    If LP_Sol is INFEASIBLE: Continue (Prune by Infeasibility);
    If Z_LP ≤ Best_Z: Continue (Prune by Bound)
    If Is_Integer(LP_Sol):
        Best_Z = Z_LP; Continue (Prune by Integrality);
    Else (Not Integer):
        x_i = Select_Fractional_Var(LP_Sol)
        v = Value(x_i);
        N1 = Current + Constraint(x_i ≤ floor(v))
        N2 = Current + Constraint(x_i ≥ ceil(v))
        Add N1, N2 to Active_Nodes;
```


Branch and Bound: Definition and Knapsack Example

Branch and Bound: An Exact Algorithm

- An exact algorithm that explores the solution space by:
 1. **Branching:** Partitioning the feasible region into smaller subproblems.
 2. **Bounding:** Calculating upper (maximization) or lower (minimization) bounds on the optimal value for each subproblem.
 3. **Fathoming (Pruning):** Discarding subproblems that cannot contain the optimal solution (e.g., bound is worse than the current best solution).

Knapsack Problem Example (Binary)

Problem: Maximize Value, subject to Capacity.

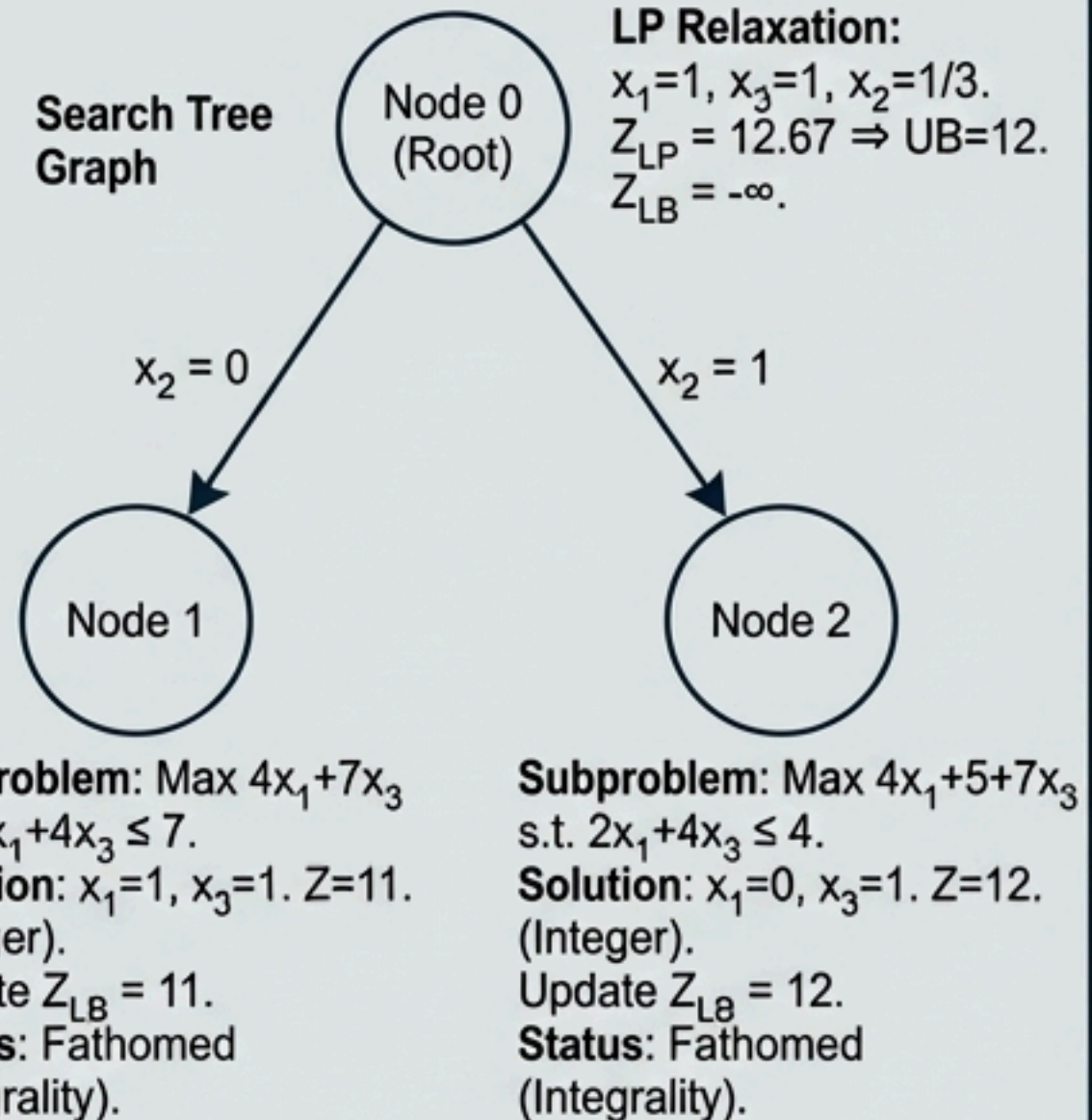
Formulation:

Max $Z = 4x_1 + 5x_2 + 7x_3$
(Values)

Subject to $2x_1 + 3x_2 + 4x_3 \leq 7$
(Weights)

$x_i \in \{0, 1\}$ (Binary decision)

Search Tree Diagram



Conclusion: Optimal Solution: $x^* = (0, 1, 1), Z^* = 12.$
All nodes fathomed.

Conclusion: The Mechanics of Exact Optimization

Bounds Dictate Limits

Primal bounds define feasible reality, while dual bounds map theoretical limits; optimality is strictly certified when their gap is zero.

Relaxation Enables Computation

LP, Lagrangian, and Surrogate relaxations temporarily sacrifice integer accuracy to rapidly compute valid dual bounds.

Duality is Symmetric

Strong Duality and Complementary Slackness theorems mathematically guarantee that primal and dual problem spaces mirror one another at optimality.

Branch-and-Bound is Definitive

By unifying branching (division), bounding (relaxation), and fathoming (pruning), B&B ensures an exact integer optimum in finite time.

Exact Approaches

Exact optimization methods, such as as Branch-and-Bound and Integer Linear Programming, systematically explore the entire solution space, using mathematical rigor to guarantee finding the global optimal solution, as demonstrated by the algorithms and examples in this presentation.